

Chapter 2

Integrating evidence into shared decision making with patients

with STEVEN J KAMPER and CATHERINE SHERRINGTON

Incredibly, some scientifically educated medical school graduates still see evidence as inferior to intuition and experience: those are the ones who give me chills.

Dr Harriet Hall

Case: introduction

Your patient is Mrs J, a 55-year-old woman with knee pain. She complains of pain, stiffness and mechanical symptoms consistent with a meniscal tear. She has had an X-ray in which the radiologist notes degenerative changes consistent with osteoarthritis.

Question

Should you recommend surgical or non-surgical management?

to junior clinicians and personal clinical experience weighed more heavily than quality research evidence. We acknowledge that many clinicians provided wonderful evidence-based practice before the term 'evidence-based practice' was coined.

Today, healthcare systems and consumers expect practice to be grounded in science. This mirrors the progression to today's technology-supported world from a society based on semi-mystical beliefs and paternalism. EBP, however, asks more than just acceptance of a scientific world view; it requires explicit understanding and integration of the three elements.

WHAT IS EVIDENCE-BASED PRACTICE?

Evidence-based practice (EBP), also known as evidence-based medicine (EBM), is the dominant paradigm under which healthcare professionals across the world are now expected to practice. We use the term 'practice' to avoid the ambiguity in the word 'medicine' (which is sometimes used to refer only to the work of medical doctors). Clinicians and researchers from McMaster University in Canada in the 1980s and 90s are credited with developing the ideas behind EBP and championing the movement. The most commonly used definition comes from David Sackett¹ and it highlights the integration of three elements: best available evidence from well-conducted and relevant scientific research, clinical expertise, and patient values and preferences² (Fig. 2.1).

Evidence-based practice should provide better patient outcomes than so-called 'eminence-based practice' whereby the opinion of the clinician alone defines an optimal management decision. In the caricature of eminence-based practice, inappropriate and outdated methods and practices were perpetuated from senior

'Evidence-based practice' is the integration of best research evidence with clinical expertise and patient values—David Sackett

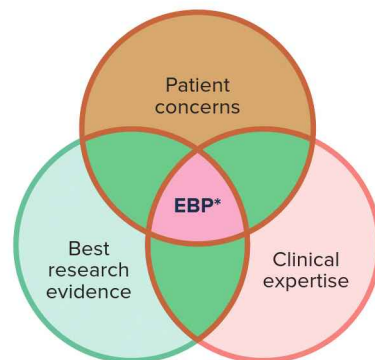


Figure 2.1 Schematic illustration of how clinical skills, evidence from research and patient desire should overlap to provide the 'quality decision' for the patient
*evidence-based practice

WHY IS THE EVIDENCE-BASED PRACTICE CONCEPT CONTROVERSIAL?

Superficially, the principles of EBP are straightforward and few argue that it represents a superior model to a paternalistic opinion-based medical model. In general, clinicians have positive attitudes towards EBP. Yet debate over the merit of EBP occupies space in numerous high-profile medical journals. In particular, there is contention when research evidence conflicts with clinical opinion that is based on observation and recall of patient outcomes.

PRACTICE PEARL

To accept the primacy of relevant high-quality research evidence requires acceptance of the flawed nature of one's own observations and recollections.

Also, engagement with EBP requires the clinician to be continually up-to-date as new scientific evidence emerges. This represents a substantial, perpetual commitment when compared to the alternative of basing decisions only on one's clinical expertise and prior academic training.

Many critics of EBP accuse its practitioners of focusing only on research results. This is an inappropriate criticism of EBP because EBP by definition integrates the best available research evidence *with* clinical expertise to guide treatment of individual patients.

PRACTICE PEARL

All three elements of EBP (not one alone), guide the final informed and shared decision about the treatment course.

CHALLENGES TO EBP

Four barriers facing clinicians who embrace EBP are:

- lack of access to digestible evidence
- inadequate skills to appraise and interpret published research^{3, 4}
- limited patient understanding of clinical issues—health literacy
- lack of time to address the above.

A lack of access to research, inadequate computer and internet services, and the inability to locate and retrieve

Case: eminence-based practice

You ask a senior colleague for advice regarding Mrs J. He has seen lots of these cases and says that Mrs J. needs to get her meniscus sorted out with an arthroscopy, the sooner the better.

articles may become less problematic with time. Clinicians' self-perceived inability to locate, appraise, synthesise and integrate research into practice reflects limitations in their training. As a newcomer to the curriculum, EBP is squeezed in behind the established curriculum. Of course, it should be naturally integrated into all clinical teaching and supplemented with formal training in technical issues such as performing or interpreting meta-analysis. As with all elements of clinical practice, 'time' limits clinicians in accessing evidence, synthesising evidence to plan best practice, and in sharing all the evidence with the patient in a 10, 20 or even 30-minute consultation. We stress that 'shared decision making' means that patient scheduling must include time for the patient to truly engage, consider and discuss the options.

IMPLEMENTING EBP

Here we outline some first steps and resources^{2, 5, 6} for clinicians aiming to follow EBP guidelines.

Accessing research

Continuously updated and freely available databases including the Physiotherapy Evidence Database (PEDro), PubMed and the Cochrane Library index health and medical literature. Their search interfaces are intuitive. These resources can be accessed via personal computers as well as tablet and mobile devices. Designing and carrying out searches of these databases is a particular skill in itself, but one that is easily learned using these free websites and their 'help' resources.

The PubMed database cites more than 24 million references in all areas of biomedicine and it provides free access to abstracts. Its 'clinical queries' feature can greatly assist in locating relevant articles (www.ncbi.nlm.nih.gov/pubmed/clinical).

The Physiotherapy Evidence Database (PEDro) indexes more than 30 000 clinical practice guidelines, systematic reviews and randomised controlled trials (RCTs) relevant to physiotherapy.⁶ Two important features of PEDro are its specificity to physiotherapy research and study design, which means that searches are more likely to find relevant studies and the methodological quality rating of all RCTs via the PEDro scale.

The Cochrane Library houses carefully conducted systematic reviews that provide the most robust evidence related to the effectiveness of interventions and diagnostic tests. Cochrane reviews are all conducted according to rigorous standards and processes to ensure maximum reliability of results. Cochrane reviews are freely accessible from all IP addresses within the countries that pay a subscription fee, as well as in developing countries.

Retrieving articles

While databases such as PEDro and PubMed provide the article abstract, this is rarely enough to appraise the reliability and applicability of a study so it is necessary to access the full text. A recent change in scientific publishing toward the 'open-access model', comes with a trend toward increasing availability of free, full-text versions of articles. This is likely to increase in the future. Currently, research articles are freely available from large general medical journals such as the *British Medical Journal* and specialty journals published by the BioMed Central (BMC) and Public Library of Science (PLOS) groups, among others. Some other individual journals make selected content or material greater than one year old freely available. An index of freely available journals can be found at www.freemedicaljournals.com. Clinicians working in universities or large hospitals will have access to the full text articles of journals for which their institution pays a subscription fee and some professional associations pay subscription fees to relevant journals so that content is available to members.

Published appraisals

In addition to complete reports of original research studies, some journals publish short summaries, including critical appraisal of studies published elsewhere. These include the *British Journal of Sports Medicine*, *Evidence-based Medicine* and the *Journal of Physiotherapy*. The Cochrane Library also includes lay-language summaries of their systematic reviews. The advantage of these short summaries is that the quality of the study and reliability of the evidence is appraised and reported along with the findings themselves.

Interpreting research about treatment effects

The top level of evidence for questions regarding the effectiveness of an intervention comes from a systematic review of RCTs; evidence from an individual RCT occupies the level underneath (Fig. 2.2). Systematic reviews of RCTs provide the highest level of evidence^{1, 2, 5} because a well-designed and conducted RCT design overcomes many of the sources of bias associated with other study types; appropriate synthesis of the findings of all available RCTs relevant to a particular question gives the most accurate estimate of the size of the treatment effect. However, the strength of the evidence depends on the number and quality of the RCTs in the systematic review.

Why prioritise RCTs over case series or observational studies when it comes to determining the effect of treatment? The key issue with observational studies is that

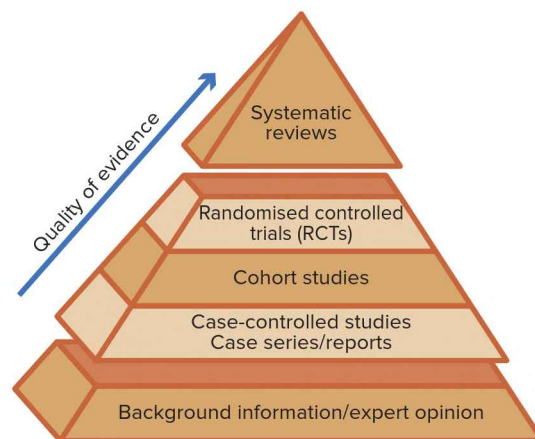


Figure 2.2 Hierarchy of evidence about the effects of treatment

we do not know what would have happened to patients if they received a different treatment or no treatment at all. It is also not enough to compare the response to treatment in one study with the response to another treatment (or no treatment) in another study, as there may be important unmeasured differences between the two samples.

Risk of bias

All research studies are not created equal; some are more likely to give a reliable answer to the question than others.^{5, 6} All RCTs indexed on the PEDro database are rated for

Case: initial search

A search of the Cochrane Library using the words 'knee and arthritis' returns 33 records. One is a Cochrane systematic review titled 'Arthroscopic debridement for knee osteoarthritis' published in 2008. The authors conclude that debridement is no different to sham surgery (based on one RCT) and no different to arthroscopic lavage (two RCTs).

Case: second search

You recognise that the systematic review is a bit old and not completely specific to patients with meniscal pathology, so you look further. A search of PEDro using the keywords 'arthritis and surgery' specifying the method 'clinical trial' and the body part 'lower leg or knee' returns 10 studies. One is an RCT titled 'Surgery versus physical therapy for a meniscal tear and osteoarthritis', from 2013. The authors found that there was no difference between the meniscectomy group and the physiotherapy group in terms of pain and disability at 6 months.

methodological quality. In practice, this means that trained raters have gone through every trial and scored whether it satisfies each of a list of criteria (the PEDro scale) that are reported on the PEDro website. (www.pedro.org.au).

The Cochrane risk of bias tool has many commonalities with the PEDro scale and is used to assess the quality of trials in many Cochrane reviews. Each item of the quality scale relates to an issue that could result in biased study results; for each criterion that is not met, the risk that the study results are biased is greater. Examples of items that might bias a study result include:

1. whether the researcher deciding whether patients were included in the study knew to which group they would be assigned (concealed allocation)
2. whether the intervention and control groups were comparable at baseline
3. whether a large proportion of the patients who began in the study remained until follow-up was complete.

Research about other types of clinical questions

Not all clinical questions relate to the effectiveness of treatments. For example, there are also relevant clinical questions about predicting the likelihood of good or poor outcomes in a patient or about the accuracy of a diagnostic test. These types of questions are fundamentally different to those about the effect of treatment, so it makes sense that the study design is different.

Questions about prediction or risk require that investigators recruit a large group of people who are at the same stage of the disease process. These participants are then followed over time to see who does well and who does not (i.e. a prospective cohort study). The investigators need to measure all the factors that might be relevant in determining the patients' future outcome (e.g. body weight in people with osteoarthritis) and make sure not too many of the participants drop out over the course of the study. A prospective cohort study can be useful to help a clinician provide evidence-based advice to patients about what to expect regarding their recovery.

Questions about diagnosis should be addressed using a study that compares the results of a diagnostic test

Case: appraisal

Looking at the PEDro scale score for the RCT you find that six of the 10 criteria were met. This indicates moderate quality, but there is the potential for bias due to certain factors, including the lack of blinding of outcome assessors and the fact that randomisation was not concealed. Further, a significant proportion of the patients randomised to the physical therapy intervention later crossed over and received surgery.

Case: decision

Based on the evidence from the review and the RCT, you are doubtful of the benefits of surgery for Mrs J. You note, however, the moderate quality of the RCT and the issue of crossover and following discussion with Mrs J decide to trial a 2–3 month program of physical therapy prior to reviewing her condition.

(the index test) with the results of an accepted standard for the diagnosis (the reference standard or gold standard), at the same time. For example, a study might compare Lachman's test for anterior cruciate rupture (detailed in Chapter 35) with MRI assessment (reference standard) in the same patients. The accuracy of the reference standard is key to the quality of a diagnostic study.

A second critical design feature of diagnostic studies is that the person performing the index test does not know the results of the reference standard. If the tester knows whether or not the patient has the condition it may bias their interpretation of the index test. So in the example, the person performing the Lachman's test should not know the results of the MRI assessment.

A final important feature concerns how the patients in the study sample are assembled. Ideally, all the patients in the sample should represent those that present for care, which means that diagnosis is unknown before the test is performed. The Cochrane Library now includes systematic reviews of diagnostic test studies.

REFERENCES

References for this chapter can be found at www.mhhe.com/au/CSM5e